INSTRUCTION MANUAL

AIR DRIVEN HIGH PRESSURE PUMP

MODEL : AHP-2500

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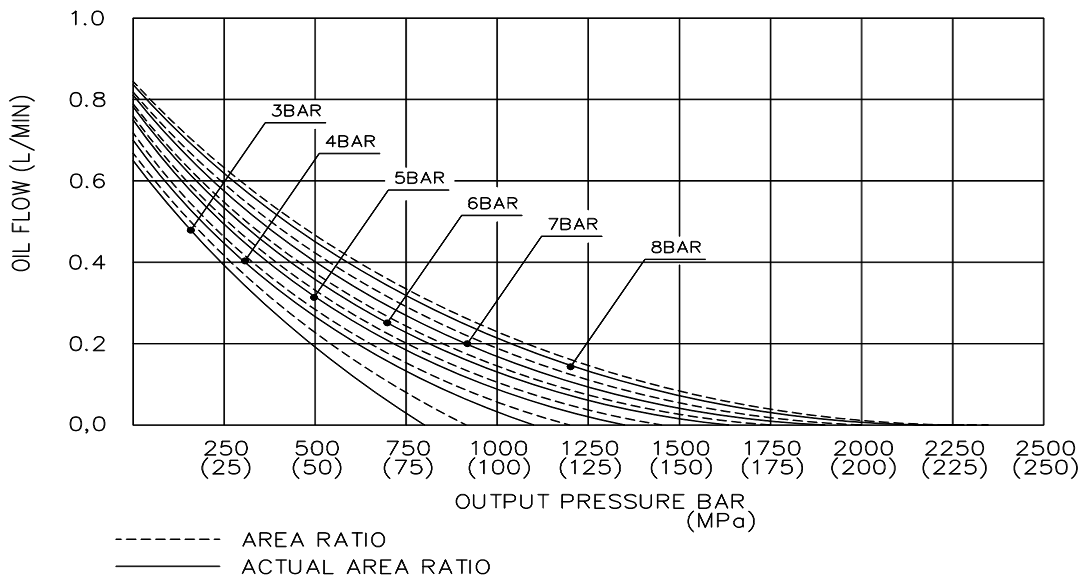
CONTENTS

APPENDIX

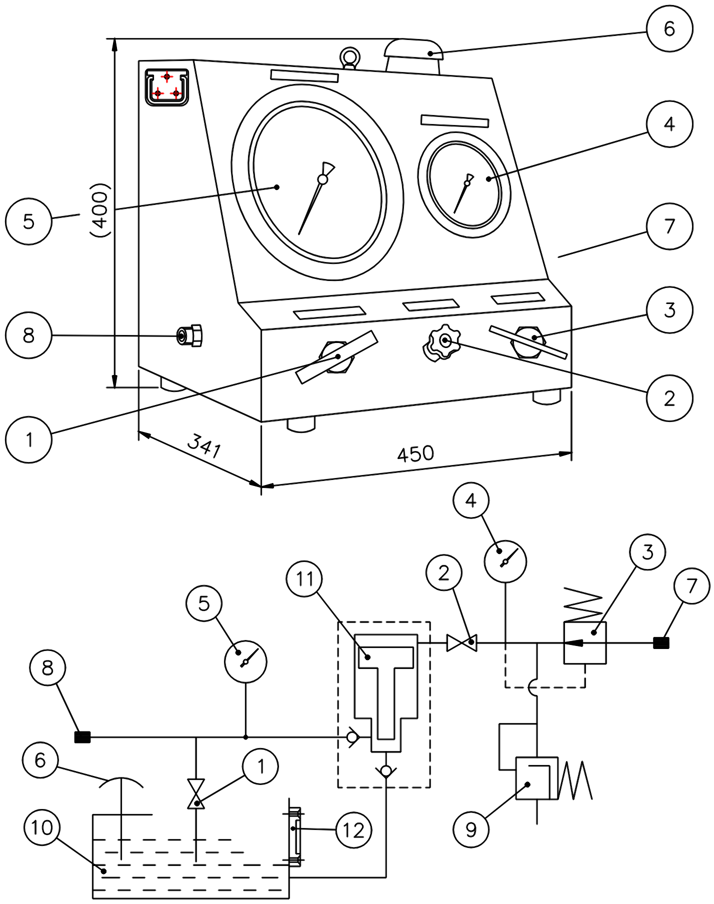
A. SPECIFICATION

1. MODEL NO.	: AHP-2500
2. AREA RATIO	: 277 : 1
3. ACTUAL AREA RATIO	: 266 : 1
4. AIR CONSUMPTION	: 3200 LITERS/MIN. AT 7 BAR AIR INPUT
5. PUMPING SPEED	: APPROX. 255 CYCLES/MIN. AT 7 BAR AIR INPUT
6. DISPLACEMENT PER CYCLE	: 3.18 ㎤
7. MAX. FLOW AT N0 PRESSURE	: APPROX. 0.81 LITERS/MIN. AT 7 BAR AIR INPUT
8. TANK CAPACITY	: 6.5 LITER
9. AIR INNER PORT	: 1/2" BSP FEMALE WITH INCREASE TO M42 ADAPTER
10. LIQUID OUTLET PORT	: 1/4" MALE
11. APPROX. DIMENSION	: 450 x 341 x 400
12. WEIGHT	: 31 Kg

PERFORMANCE CURVES



B. OUTLINE OF THE AHP-2500



1. OIL RETURN VALVE
2. AIR STOP VALVE (CYCLING VALVE)
3. AIR PRESSURE CONTROL VALVE
4. AIR PRESSURE GAUGE FOR PRESSURE CONTROL
5. OIL PRESSURE GAUGE FOR WORKING PRESSURE
6. AIR BREATHER (FILLER CAP)
7. AIR INLET (FEMALE COUPLING FOR AIR HOSE)
8. OIL OUTLET MALE COUPLING (HIGH PRESSURE OUTLET)
9. SAFETY VALVE (RELEASE VALVE)
10. OIL TANK
11. PUMP
12. OIL LEVEL GAUGE

C. OPERATING MANUAL

This instruction has been made to facilitate the understanding of the working principles of the pump unit.
The various control instruments and maneuvering handle are fitted with a number.
These numbers are mentioned in this instruction and are also referring to the numbers found on the drawings showing the complete pump unit.
Furthermore, these numbers are found on the instrument panel of the pump unit.
Please, read this direction of use carefully before using the pump unit and follow the instructions given vary carefully.

1. BEFORE CONNECTING THE PUMP UNIT

The tank is filled with oil through the filler cap ⑥ in the right side of the top panel.
Hydraulic oil : ISO VG22 OF ISO VG32 (MINERAL OIL)

a) Open oil return valve ① by turning anti-clockwise.
b) The handle of the regulator valve ③ should be turned 4-5 turns anti-clockwise.
This is done to avoid the pump starting at a too high pressure level.
c) Stop valve ② should be turned clockwise, to close the valve.
This is done to ensure that the pump will not work at will when the primary pneumatic system is connected.

2. CONNECTION OF PUMP

Compressed air is led through a flexible hose and is connected to the stud marked "air inlet" ⑦ on the right side of the cabinet. This stud has a 1/2" BSP female thread with a 1/4" BSP reduction, accepting either 1/2" or 1/4" hose. To avoid contamination of the pump, it is essential that an effective air filter and water separator is inserted in the air line. A lubro unit should however be avoided - see lubrication instruction on page 4.
The hydraulic system is connected to the stud marked "high pressure outlet" (8) found on the left side of the cabinet. This stud has a 1/4" BSP female thread.
The pump is now ready for use.

3. START OF PUMP

a) Stop valve ② is turned slowly anti-clockwise, whereby compressed air enters into the pump unit, which commences to work. The stop valve ② is also acting as a regulator valve for the pumping speed.
b) The regulator valve ③, which is used for adjusting the pressure is now turned clockwise and the hydraulic pressure may now be read on the manometer for pressure control ④
This manometer shows the hydraulic high pressure in bar.

c) The oil return valve ① is closed by turning the knob clockwise and the oil will then run from the oil tank into the hydraulic system. When the hydraulic system has been filled. and the pump has stopped operating, the high pressure reached can be read at the manometer for working pressure ⑤ found on the left side of the instrument panel.
The pump stops automatically, when the required high pressure has been eatablished and holds it infinitely.
The pump starts automatically again if a pressure drop occurs in the hydraulic system.

d) High pressure may be relieved from the system by opening the oil return valve ① (to be turned anti-clockwise). The excessive oil the secondary high pressure system will thereby return to the oil tank in the pump unit.

4. ENDING OF JOB

To avoid oil spillage, precautions should be taken to see that the oil is taken back into the oil tank before the connections on the high pressure side are disconnected.
This is done by turning the oil return valve ① anti-clockwise.
Simultaneously the pump should be stopped by turning the stop valve ② clockwise, and when this valve has been closed, the air hose may be removed and the pump transferred to another job.

5. DATA FOR HIGH PRESSURE

The pump unit can deliver a maximum pressure of 1500 bar. At an air pressure of 4 bar a safety release valve is in the B+W standard version adjusted to release at a maximum pressure of 1060 bar.
This pressure of 1060 bar (which is preset by the factory) secures the rest of the system from overload or faulty operation. The pump may be re-adjusted to blow off at any desired pressure, but maximum 1500 bar.
The release valve should be set at approx. 10% above the desired working pressure.

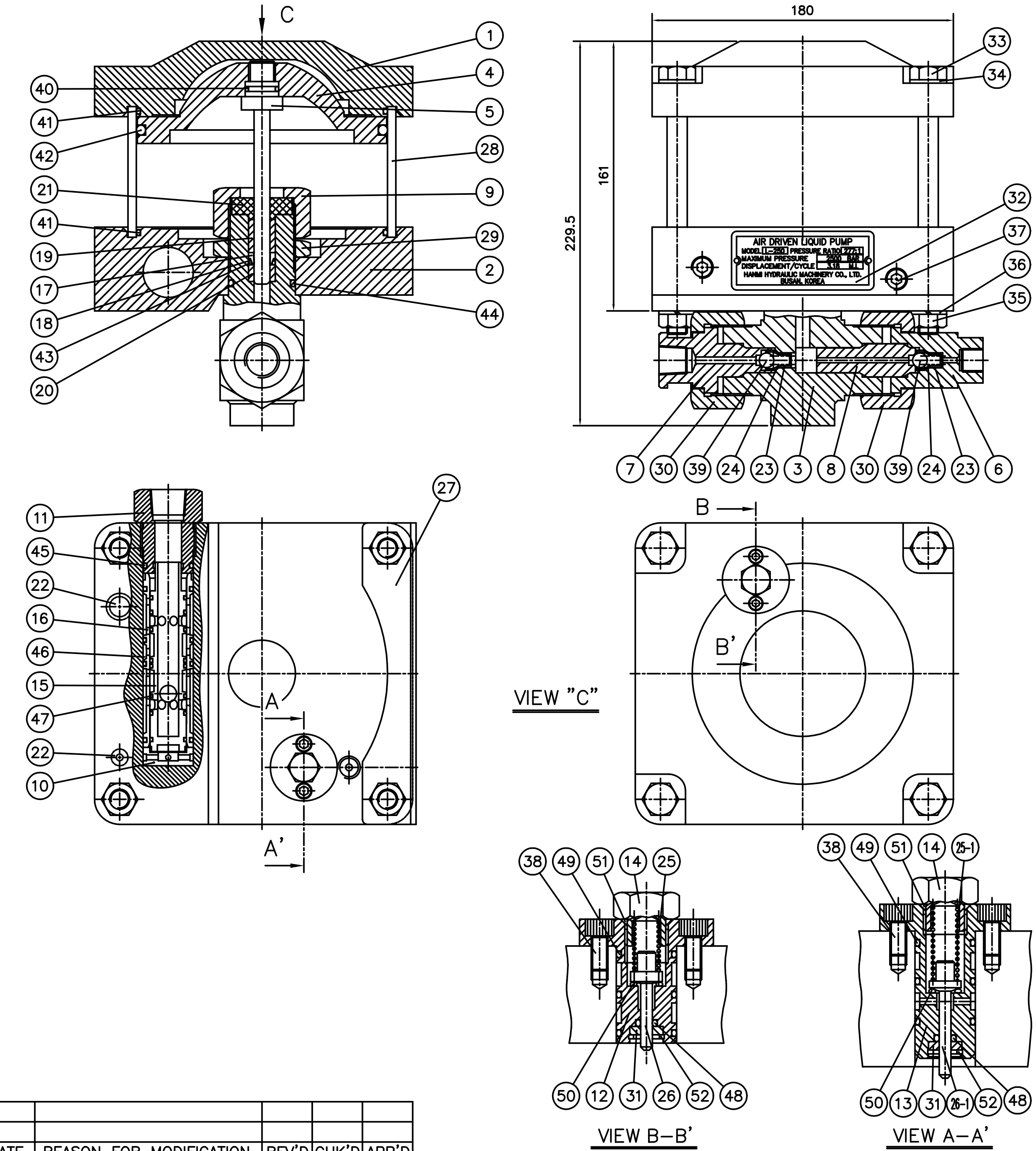
6. CLEANING OF OIL FILTER

The filter in the filler cap ⑥ is removed and cleaned when by re-filling the oil starts running through slowly.

7. LUBRICATION INSTRUCTION

WARNING : AVOID AIR LINE LUBRICATION.

The pump has been pre-lubricated and a lubro unit will decrease the efficiency of the pump.
Air valves should be lubricated periodically.
Remove exhaust fittings and valve spool and apply a thin coat of hydraulic grease.
If primary airdrive section is dismantled, always apply a thin coat of hydraulic grease to entire inner surface of cylinder tube plunger O-rings.



NO.	DESCRIPTION	DWG. NO.	Q'TY	REMARK
52	SNAP RING	10–2152–0	2	
51	O–RING	10–2151–0	2	
50	O–RING	10–2150–0	2	
49	O–RING	10–2149–0	6	
48	O–RING	10–2148–0	2	
47	O–RING	10–2147–0	8	
46	O–RING	10–2146–0	4	
45	O–RING	10–2145–0	1	
44	O–RING	10–2144–0	1	
43	O–RING	10–2143–0	1	
42	O–RING	10–2142–0	1	
41	O–RING	10–2141–0	2	
40	O–RING	10–2140–0	1	
39	BALL	10–2139–0	2	
38	WRENCH BOLT	10–2138–0	4	
37	PLUG	10–2137–0	11	
36	HEX. NUT	10–2136–0	4	
35	SPRING WASHER	10–2135–0	4	
34	WASHER	10–2134–0	4	
33	HEX. BOLT	10–2133–0	4	
32	NAME PLATE	10–2132–0	1	
31	RING	10–2131–0	2	
30	GLAND NUT	10–2130–0	2	
29	NUT	10–2129–0	1	
28	AIR CYLINDER	10–2128–0	1	
27	PUMP BRACKET	10–2127–0	1	
26–1	PILOT STEM	10–2126–1	1	
26	PILOT STEM	10–2126–0	1	
25–1	SPRING	10–2125–1	1	
25	SPRING	10–2125–0	1	
24	SPRING	10–2124–0	2	
23	RETAINER	10–2123–0	2	
22	FLOW TUBE	10–2122–0	1	
21	FLANGE	10–2121–0	1	
20	RING	10–2120–0	1	
19	RING	10–2119–0	1	
18	RING	10–2118–0	1	
17	PACKING SEAL	10–2117–0	1	
16	SLEEVE	10–2116–0	1	
15	SPOOL	10–2115–0	1	
14	PLUG	10–2114–0	2	
13	BOTTOM V/V BODY	10–2113–0	1	
12	TOP V/V BODY	10–2112–0	1	
11	RETAINER	10–2111–0	1	
10	RETAINER	10–2110–0	1	
9	GLAND NUT	10–2109–0	1	
8	SEAT	10–2108–0	1	
7	FITTING	10–2107–0	1	
6	FITTING	10–2106–0	1	
5	PISTON	10–2105–0	1	
4	AIR PISTON	10–2104–0	1	
3	H.P CYLINDER	10–2103–0	1	
2	BOTTOM CAP	10–2102–0	1	
1	TOP CAP	10–2101–0	1	

APPROVED Y.K.JUN	SCALE N / S	PROJECT AIR DRIVEN HYD PUMP–L(LS)TYPE	PROJECTION
CHECKED M.H.CHOI	DATE 2002.8.14	TITLE ASSEMBLY OF BOOSTER	
DESIGNED D.H.KOO		HANMI HYDRAULIC MACHINERY CO., LTD.	DRAWING NO. M10–2100–0.0

REV.	DATE	REASON FOR MODIFICATION	REV'D	CHK'D	APP'D
△					
△					

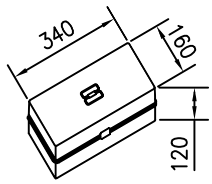
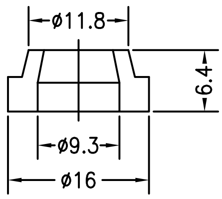
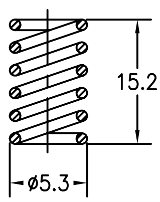
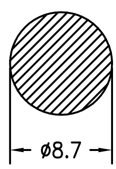
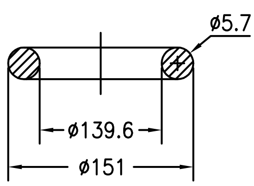
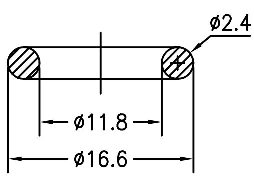
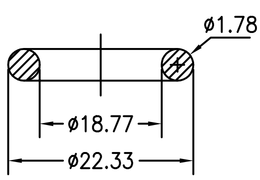
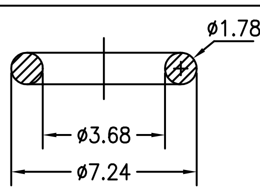
SPARE

AIR DRIVEN HIGH PRESSURE PUMP

MODEL : AHP-2500

PAGE 7

DWG. NO. M10-2107-0.0

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SHIP STAND-ARD	SUPPLY PER SHIP ADDIT-ION	SUPPLY PER SHIP TOTAL	DRAWING NO.	REMARK (WEIGHT) Kg
1	BOX		PLASTIC	1	–	1		
1-1	PACKING SEAL		UHMWPE	1	–	1	10-2117 -0	
1-2	SPRING		SUS304	2	–	2	10-2124 -0	
1-3	BALL		STEEL	2	–	2	10-2139 -0	
1-4	O-RING		N.B.R	1	–	1	10-2142 -0	
1-5	O-RING		N.B.R	1	–	1	10-2143 -0	
1-6	O-RING		N.B.R	8	–	8	10-2147 -0	
1-7	O-RING		N.B.R	2	–	2	10-2148 -0	

MANUFACTURER'S NAME	HANMI HYDRAULIC MACHINERY CO., LTD.